Analysis and Prediction of Criminal Activities in Los Angeles

Miljana Marjanović E2 99/2024
Milena Marković E2 100/2024

1. Problem Definition

The goal is to collect and analyze data on criminal activities in Los Angeles, along with weather conditions and demographic information from different parts of the city. Special focus will be placed on exploring potential correlations between criminal activities and the aforementioned additional factors. After data processing and analysis, various models and algorithms will be tested in order to predict criminal activities in the Los Angeles area.

2. Problem Motivation

By analyzing criminal activities, certain patterns may be identified, providing law enforcement officers with a clearer overview of the situation on the ground. This approach offers insights into high-risk zones and can assist in planning police patrol schedules. Taking into account demographic data and weather conditions, such analysis can enable faster and more efficient responses from the authorities, thereby directly contributing to the reduction of crime rates.

3. Relevant Literature

3.1 Exploratory data analysis of crime report, 2021 [pdf]

Topic: This paper focuses on the analysis and visualization of crime activity data using the R programming language. The charts presented illustrate the distribution of crimes by year, month, day, gender, and other attributes.

Data: The study uses crime activity data from Los Angeles covering the period from 2010 to 2017. [link] The dataset consists of 2,000,000 records with the following attributes: date, time, and exact location of the crime, report date, crime type, weapon usage, age and gender of the victim, and more.

Methodology: Since the paper is centered on data analysis, methodology is not its primary focus. It employs standard techniques such as variable identification, data preprocessing, and univariate, bivariate, and multivariate analysis. Data visualization is done using the ggplot2 package.

Conclusion: We will use the dataset from this study and extend it with additional information on weather conditions and demographic characteristics. The same analytical techniques from the referenced paper will then be applied to uncover new patterns and gain a deeper understanding of the factors influencing crime.

3.2 Risk Prediction and Severity Classification of Motor Vehicle Collisions in New York, 2021 [pdf]

Topic: This study analyzes the factors that directly influence the likelihood of motor vehicle collisions (e.g., collision data, weather conditions, traffic flow). It also focuses on predicting collisions and classifying their severity.

Data: Three publicly available datasets from NYC Open Data were used [link], containing information about collisions and traffic flow on individual street segments. Weather data was retrieved via the IBM Weather API [link].

Data Preprocessing: Since location was a key feature for merging the three datasets, all entries lacking this attribute were removed. The collision data was merged with weather information based on the date and hour of the incident. Due to the presence of cyclical date and time features (e.g., 11 PM and 12 AM), sine and cosine transformations were applied. One-hot encoding was used for categorical features. To enable predictive modeling, a negative sample was generated for each data point — with the same date but a randomly chosen street and time, under the condition that no collision occurred at that place and time. The dataset was split into training (90%) and test (10%) sets.

Methodology: The prediction task was treated as a binary classification problem (whether a collision occurred or not). Tree-based models were used, including Random Forest, Extra Trees, LightGBM, XGBoost, and CatBoost, along with the K-Nearest Neighbors (KNN) model.

Evaluation: Model performance was evaluated using precision, accuracy, recall, and F1 score. CatBoost achieved the best performance with an F1 score of 0.7.

Conclusion: We will adopt the data preprocessing approach from this study, applying the same cleaning and merging techniques to our datasets on crime activities, weather, and demographics. Additionally, we will implement the listed prediction models and compare their performance to determine the best fit for our dataset. The same evaluation metrics will be used for consistency.

3.3 A Crime Data Analysis of Prediction Based on Classification Approaches, 2022 [pdf]

Topic: This paper focuses on predicting the type of criminal activity in Boston. The main goal is to compare the performance of three supervised learning algorithms: Decision Tree, Naïve Bayes, and Logistic Regression.

Data: The analysis is based on data from the Boston crime incident report database, covering the period from mid-2015 to the first half of 2018. [link] The dataset classifies the type of incident and includes detailed information about the time and location of each event.

Data Preprocessing: Missing values were filled using the mean value of the respective attribute. Afterward, the data was normalized to a range of 0 to 1. The dataset was then split into training (80%) and test (20%) sets for model training and evaluation.

Methodology: The goal was to predict the type of criminal activity. The following algorithms were used: Decision Tree, Naïve Bayes, and Logistic Regression. Among them, the Decision Tree algorithm produced the best results.

Evaluation: The metrics used for evaluation were precision, recall, and F1 score. Decision Tree achieved the highest performance, scoring 1.00 across all three metrics.

Conclusion: Our plan is to implement these algorithms and compare their performance with the models from the previous study (Random Forest, Extra Trees, LightGBM, XGBoost, and CatBoost). This comparison will help us determine which model delivers the most accurate and reliable results for our dataset.

4. Dataset Description

Crime Data [link]

As the primary dataset, we will use crime data from Los Angeles covering the period from 2010 to 2019. This dataset includes over 2,000,000 records and contains attributes such as the exact time and location of the incident, as well as the type of crime committed. In addition to these core details, the dataset provides supplementary attributes such as the age, gender, and race of the victim, whether a weapon was involved, and the specific nature of the location (e.g., whether the incident occurred at a restaurant, on a parking lot, or on the street). All of these attributes will be analyzed in detail to identify key factors that will serve as a foundation for further processing and investigation.

Socio-Economic Data [link]

Los Angeles is divided into 15 districts, and for each of these, there are population census data available from 2010. These data include various population distributions, covering information such as race, age, gender, and other key demographic factors. We will use the 2010 census data under the assumption that the demographic situation did not change significantly from 2010 to 2019. Since data from the 2020 census have not yet been published on the same platform, if we find a suitable dataset, we will use an approximation based on the demographic information from both the 2010 and 2020 censuses.

Weather Data [link] **or** [link2]

Weather data will be retrieved from one of the two proposed sources. The attributes considered will include temperature, precipitation, humidity, wind speed, and other relevant factors. Based on our analysis, we will determine which attributes have the most significant impact, and only those will be included in further work.

5. Methodology

After preprocessing all datasets, we will merge relevant information on criminal activities, weather conditions, and demographic characteristics into a unified dataset. This consolidated dataset will serve as the foundation for further analysis. Once merged, we will generate a corresponding negative example for each crime record. The negative sample will retain the same date but will feature a randomly selected location, under the condition that no criminal incident occurred at that place and time. All samples will then be labeled as positive (actual crimes) or negative (no crimes). We will then evaluate most of the mentioned methods for predicting criminal activities, including Random Forest, Extra Trees, LightGBM, XGBoost, CatBoost, Decision Tree, Naïve Bayes, and Logistic Regression. Based on the performance of each model, we will determine which algorithm delivers the best results.

6. Evaluation Method

The prediction models will be evaluated using the following metrics: precision, accuracy, recall, and F1 score. The dataset will be split into training, validation, and test sets with an 80/10/10 ratio.